

You’re Pushing My Buttons: Instrumented Learning of Gentle Button Presses

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Abstract—Contact-rich robotic manipulation tasks remain challenging for vision-only control policies because of occlusions and indirect physical feedback. In this work, button pressing is used as a controlled example application to study an instrumentation-centric learning pipeline, rather than as a difficult end task by itself. We sensorise the environment and use the button state as privileged supervision to shape task-relevant acoustic representations from a generic Audio Spectrogram Transformer. We then transfer this instrumented signal into a diffusion policy through three integration strategies, while requiring only vision and audio at inference time. Experiments show that success rates are similar across methods, but instrumentation-guided audio features substantially regularise contact forces.

Index Terms—Robotic Manipulation, Audio-Tactile Integration, Diffusion Policies, Instrumentation

I. INTRODUCTION AND RELATED WORK

Contact-rich manipulation tasks, such as button pressing, require tight force control. Vision-only policies often fail in this setting because end-effector contact is partially occluded and visual observations provide limited information about impact dynamics. Recent work shows that audio can supply this missing contact signal. ManiWav [1] integrates an AST-based [2] audio stream and reports better manipulation performance when the audio encoder is trained from scratch on task audio than when using generic AudioSet-pretrained features [3], suggesting that contact-specific spectra are not well captured by broad ambient-audio pretraining. Simultaneously, we have proposed the use of instrumentation, i.e., integration of sensors in the robot’s environment, as a powerful tool for robot learning. In previous work, we show that instrumentation can enable automation of high-quality demonstrations [4], and that by including privileged instrumentation data in the state observation, policy performance can be improved and distilled to a vision-only policy [5]. Similarly [6] and [7] used instrumentation to obtain privileged information directly in the real world, which enabled the difficult manipulation tasks of raspberry harvesting and solving a Rubik’s cube, respectively.

In this paper, we use instrumentation for high-quality data collection and to fine-tune a task-specific audio encoder, while deploying policies that rely only on vision and sound at inference time. Our key research question is whether privileged real-world sensor data can improve manipulation performance during training, and which integration strategy best transfers

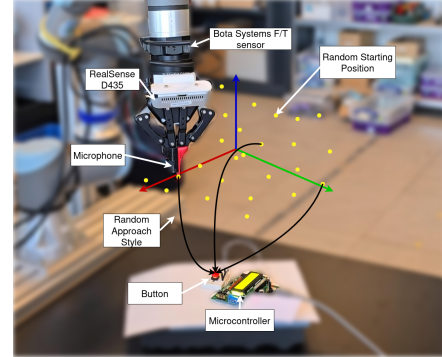


Fig. 1. Overview of the hardware setup. In addition, exemplary randomised end-effector start positions and approach trajectories are indicated.

these benefits to a policy that does not require privileged signals when deployed.

II. METHODS AND MATERIALS

Experimental Setup: The hardware setup, shown in Fig. 1, consists of a UR3e robotic arm equipped with a Robotiq 2F-85 gripper and a Bota Systems F/T sensor: BFT-DENSER-M8. Observations are captured using a wrist-mounted Intel RealSense D435 camera. To capture acoustic data, an ICS-43434 microphone is integrated in a custom 3D-printed fingertip. A push button is wired to a microcontroller that reads the binary button state (pressed or unpressed) and transmits it as privileged information.

Task Definition and Automated Data Collection: The manipulation task requires the robot to approach the button, press it, and retract. We leverage instrumentation to collect automated expert demonstrations: the end-effector (EE) position of the robot is randomly initialised, and approaches the button with randomised horizontal and vertical speeds (see approach trajectories in Fig. 1). As soon as a button press is detected, the robot retracts. This results in cleaner demonstrations than a human teleoperator can produce, similar to the approach in [4].

Observation and Action Spaces: We use Diffusion Policy [8] implemented via the LeRobot framework with default settings unless otherwise specified. To guarantee reactivity, we set `n_action_steps` to 1. Each observation at time t consists of:

- **Wrist RGB Image (I_t):** Processed by a default ResNet encoder. Images are randomly cropped to 216×288 during training and center-cropped during inference.
- **Audio Spectrogram ($A_t \in \mathbb{R}^{298 \times 128}$):** A log-Mel spectrogram computed over the most recent 3 seconds of audio.

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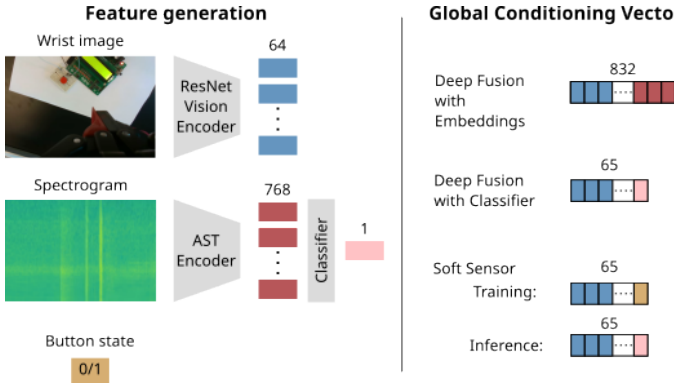


Fig. 2. Overview of the audio integration strategies evaluated in this work: Deep Fusion (direct conditioning on AST logits or embeddings) and Soft Sensor (training with true button state, replaced by AST prediction at inference time).

Actions are defined as position deltas $\mathbf{a}_t = [\Delta x, \Delta y, \Delta z]$ relative to the current EE frame. This action space renders the policy invariant to the positioning of the button.

Audio Processing and AST Fine-tuning: Contact-rich tasks, such as precisely pressing a button and safely retracting, are difficult for vision-only policies due to visual occlusion and the absence of physical feedback. To address this, we introduce acoustic data from the fingertip-mounted microphone as a proxy for tactile feedback. The 3-second audio windows are resampled to 16 kHz and converted into log-Mel spectrograms (128 bins, 25 ms window, 10 ms hop size). Initially, these spectrograms are encoded using a generic Audio Spectrogram Transformer (AST) [2], a DeiT-based architecture [9] pretrained on AudioSet [3].

However, to improve the performance of this audio-tactile policy, we leverage hardware instrumentation. By using the button signal as automated training labels, we fine-tune the generic AST to detect the acoustic signature of a successful button press. We fine-tuned the pretrained AST with a 2-class classification head for 10 epochs with learning rate 10^{-5} . Utilising an AST-specific normalisation. The model achieved an F1 score of 0.988 and a false negative rate of 1.2% on the validation set. This process effectively ingrains the privileged instrumented data into a task-specific acoustic detector.

Integration Strategies: Fig. 2 summarises the two policy variants:

- **Soft Sensor (Pseudo-Instrumentation):** train with privileged button state in the observation; replace it at inference with AST-predicted button state.
- **Deep Fusion:** inject audio during training using either (i) AST classifier logits or (ii) AST embedding features.

We compare both against a baseline that uses embeddings from a generic AST pretrained on AudioSet, without task-specific fine-tuning.

We train all policies with Adam (learning rate 10^{-4} , weight decay 10^{-6}) and a cosine schedule with 500 warmup steps. We apply standard image augmentations (brightness, contrast, saturation, hue, and sharpness jitter) and normalize actions/states to $[-1, 1]$; spectrograms use AST normalisation.

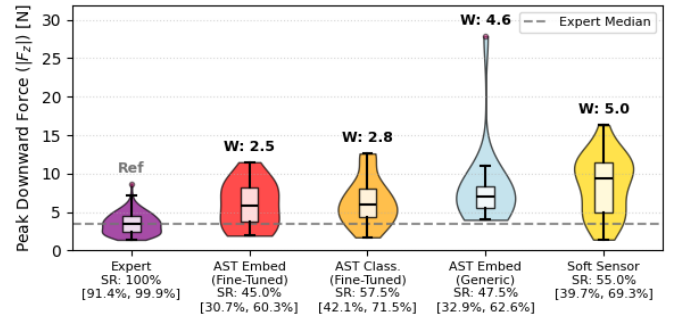


Fig. 3. Force distributions ranked by Wasserstein distance (W [N]) to the Expert demonstrations. Success rates (SR) and Bayesian 95% credible intervals are provided for each model.

Evaluation Protocol: Each policy is evaluated on 40 rollouts. Initial EE poses are uniformly randomised within the same workspace used for data collection. A rollout is “successful” if the button is pressed and the robot subsequently retracts. Additionally, wrist force-torque signals are logged to assess rollout quality: a successful press with minimal contact force is preferable to one with high force.

III. RESULTS AND DISCUSSION

As shown in Fig. 3, success rates are statistically indistinguishable at 40 rollouts per model: the Bayesian 95% credible intervals largely overlap. We therefore focus on contact-force metrics to differentiate policy quality.

We analysed peak vertical force (F_z) during successful presses. The expert demonstrations indicate optimal force levels (avg 3.54 N, median 3.41 N). The generic AST baseline applies higher forces (avg 8.12 N, median 6.98 N). Deep Fusion improves contact behavior: frozen fine-tuned embeddings yield avg 5.99 N (median 5.87 N), and the fine-tuned classifier yields avg 6.36 N (median 5.95 N). The Soft Sensor variant performs worse (avg 8.52 N, median 9.37 N).

Computing the Wasserstein distance from each distribution to the expert force distribution confirms this ranking: Soft Sensor is least similar ($W=5.0$ N), generic AST remains distant ($W=4.6$ N), and Deep Fusion with fine-tuned embeddings ($W=2.5$ N) or fine-tuned classifier ($W=2.8$ N) is closest to expert behavior. This proves that conditioning the policy on fine-tuned acoustic representations is an effective method for regularising contact forces.

IV. CONCLUSION AND FUTURE WORK

This paper frames button pressing as a minimal testbed to demonstrate a broader point: instrumentation can provide privileged supervision that teaches policies better contact behaviour, even when that privileged signal is removed at deployment. Rather than claiming that AST fine-tuning alone solves a hard task, we show how instrumentation-guided representation learning improves interaction quality by reducing excessive contact force at comparable success rates. Future work will present alternative integration strategies, such as end-to-end AST unfreezing and audio-visula fusion, and evaluate whether these gains transfer across different button mechanisms.

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